

Count-Only Inverse-Variance Weighting for Intervention Harvesting

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Abstract

Intervention-harvesting estimators of position bias [1] combine per-cell click-rate estimates across many $(q, d, \text{position})$ cells whose impression counts span several orders of magnitude. Under the Position-Based Model (PBM), the per-cell Bernoulli variance $p_k r_i (1 - p_k r_i) / N_k^i$ depends on the (unknown) cell relevance r_i ; the actual inverse-variance optimum is therefore not deployable from counts alone. We work with the *relevance-agnostic count-only surrogate* obtained by dropping the r_i -dependent factors. On this surrogate, the *harmonic* weight $\omega_i = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ is the inverse-variance optimum—Aitken’s classical generalised-least-squares solution [2] restricted to count-only weights—and an unconditional Cauchy–Schwarz inequality shows it minimises $V_k + V_{k'}$ for any impression counts; empirically the same reduction holds for the asymmetric variant $\alpha V_k + \beta V_{k'}$ on KDD ($\alpha/\beta \approx 0.50$). We propose the plug-in delta-method asymptotic variance $\tilde{V}(\omega)$ of the self-normalised weighted ratio estimator [12, 14] as a finite-sample diagnostic that is well aligned with the conditional-on-counts efficiency question, in contrast to a naive pair-bootstrap. On the full KDD Cup 2012 Track 2 sponsored-search log (235M impressions, 4.6M interventional pairs), the plug-in asymptotic relative efficiency of harmonic over uniform is 4.53, well-calibrated against click-resample standard error (1–2% at two sample fractions), and robust to weight-capping (ARE ≥ 4.1 at the 99.99th percentile). A Cochran- Q diagnostic confirms substantial PBM mis-specification ($Q/\text{df} = 3.83$), allowing efficiency comparisons to be interpreted separately from model fit; under such mis-specification different weight schemes target different weighted averages of cell-specific ratios—an estimand caveat we discuss explicitly. Synthetic (43%–54% MSE reduction at high imbalance) and Yahoo-LTR semi-synthetic ($\sim 40\%$ variance reduction at $r = 50$) experiments with ground truth corroborate the efficiency prediction.

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1 Introduction

Position bias systematically corrupts click signals used for learning-to-rank [9]; intervention harvesting [1] estimates position-bias ratios $R_{k,k'} = p_k / p_{k'}$ from naturally occurring ranking variations

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without explicit randomisation. On production click logs, the per-cell counts that feed this estimator are extraordinarily heavy-tailed. On the full KDD Cup 2012 Track 2 release (235M impressions; 4.6M interventional pairs at adjacent positions (1, 2)), **60.7% of pairs have $N_1 + N_2 \leq 5$ while 0.1% of pairs carry 42% of all impressions** [10]. Public benchmarks can hide this shape: the released Baidu-ULTR [17] aggregation has 94% of (q, d, pos) cells with $N = 1$ and 78% of adjacent-pair impression ratios within $[0.5, 2]$, which we attribute to the per-cell deduplication used in the public release rather than to the underlying production logs (the unaggregated KDD release, by contrast, preserves the heavy-tailed production distribution).

Equal-weighting heteroscedastic estimators is statistically sub-optimal by a century of classical theory: Aitken’s generalised-least-squares theorem [2] mandates inverse-variance weights, a result the meta-analysis literature [5, 8] has championed for fifty years. Recent ULTR work [1, 15] introduced *harmonic* weighting $\omega_i = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ to reduce variance versus uniform on heavy-tailed counts, but existing validation has relied on *bootstrap* resampling, which we show can be misleading for inverse-variance-style weights on singleton-dominated heavy-tailed data (Section 3).

Contributions. (1) We identify harmonic as the count-only inverse-variance choice under the standard PBM with homoscedastic relevance (Theorem 2), prove an unconditional Cauchy–Schwarz reduction of the delta-method variance *surrogate* $V_k + V_{k'}$ (Theorem 3, conditional on the surrogate objective and unconditional in the counts), and characterise the exact asymmetric surrogate oracle of which harmonic is the $\alpha = \beta$ special case (Theorem 4). (2) We adopt the plug-in delta-method asymptotic variance $\tilde{V}(\omega)$ as a finite-sample efficiency diagnostic aligned with the conditional-on-counts question, and use empirical click-resample SE as its calibration baseline; we discuss why pair-bootstrap is misaligned with this question on heavy-tailed data. (3) On the full KDD log we measure $\text{ARE}(\text{harmonic}, \text{uniform}) = 4.53$, calibrated within 1–2% across two sample fractions, and robust to weight-capping at the 99.99th percentile ($\text{ARE} \geq 4.1$). We quantify PBM mis-specification via Cochran’s $Q/\text{df} = 3.83$ and discuss the estimand caveat that follows. Synthetic and Yahoo-LTR experiments with ground truth corroborate.

2 Estimator and Theory

Under PBM, $P(\text{click} \mid q, d, k) = p_k r_{q,d}$. Given an interventional set $I_{k,k'}$ of (q, d) pairs that appeared at both positions k and k' , the position-bias ratio is estimated as [1]

$$\hat{R}_{k,k'} = \frac{A}{B} = \frac{\sum_i \omega_i \hat{\theta}_k^i}{\sum_i \omega_i \hat{\theta}_{k'}^i}, \quad \hat{\theta}_k^i = \frac{C_k^i}{N_k^i}. \quad (1)$$

This is a self-normalised weighted ratio estimator. We say a weight scheme is *ratio-unbiased* when the ratio of conditional expectations of numerator and denominator equals $p_k / p_{k'}$, i.e. $\mathbb{E}[A \mid N] / \mathbb{E}[B \mid$

$N] = p_k/p_{k'}$ (not $\mathbb{E}[\hat{R}] = p_k/p_{k'}$; the finite-sample bias of the self-normalised ratio is $O(1/N)$ under standard regularity). Under PBM, the ratio of conditional expectations of numerator and denominator equals $p_k/p_{k'}$ for any count-only weight $\omega_i = \omega(N_k^i, N_{k'}^i)$, and the resulting self-normalised ratio estimator is consistent for this target [1].

PROPOSITION 1 (CONDITIONAL DELTA-METHOD VARIANCE). *By the delta method [14], $\text{Var}(\hat{R} \mid N) \approx R^2[\alpha V_k(\omega) + \beta V_{k'}(\omega)]$ with $V_r(\omega) = \sum_i \omega_i^2 / N_r^i / (\sum_i \omega_i)^2$, $\alpha = (1 - p_k)/p_k$, $\beta = (1 - p_{k'})/p_{k'}$.*

Relevance-agnostic count-only IVW. The exact Bernoulli variance is $\text{Var}(\hat{\theta}_k^i) = p_k r_i (1 - p_k r_i) / N_k^i$, which depends on the cell relevance r_i . Plug-in inverse-variance weights with $\hat{r}_i$ estimated per cell define the θ -aware oracle of Theorem 4; in heavy-tailed click logs $\hat{r}_i$ is itself extremely noisy on the $\sim 60\%$ singleton mass, making this oracle infeasible in practice (Table 2: ARE = 0.002). We therefore work with a *relevance-agnostic surrogate* that drops the r_i -dependent factor from the cell-level variance. The inverse-variance optimum of Aitken [2] on this surrogate, restricted to count-only weights, is the harmonic mean of the two cell counts. We make this precise:

THEOREM 2 (COUNT-ONLY IVW ON THE SURROGATE). *Among all non-negative weights $\omega_i = \omega(N_k^i, N_{k'}^i)$, harmonic weighting minimises the count-only delta-method variance surrogate $V_k(\omega) + V_{k'}(\omega)$.*

THEOREM 3 (SURROGATE GUARANTEE, UNCONDITIONAL IN COUNTS). *For any impression counts $\{N_k^i, N_{k'}^i\}_{i=1}^n$, $V_k(\omega^h) + V_{k'}(\omega^h) \leq V_k(1) + V_{k'}(1)$. The guarantee is unconditional in the counts but conditional on $V_k + V_{k'}$ as the objective; it implies improved $\text{Var}(\hat{R} \mid N)$ via Proposition 1 when $\alpha \approx \beta$. For the asymmetric $\alpha V_k + \beta V_{k'}$ surrogate the bound is empirical only, but we verify on full KDD that harmonic reduces it by essentially the same factor as the symmetric one (Section 4).*

SKETCH. Substituting $\omega_i^h = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ collapses the LHS to $1/W$ with $W = \sum_i \omega_i^h$. The RHS is $(1/n^2) \sum_i 1/\omega_i^h$. Cauchy-Schwarz gives $n^2 \leq W \sum_i 1/\omega_i^h$. $\square$

THEOREM 4 (EXACT ASYMMETRIC ORACLE). $\alpha V_k(\omega) + \beta V_{k'}(\omega) \geq (\sum_i 1/s_i)^{-1}$ with $s_i = \alpha/N_k^i + \beta/N_{k'}^i$, attained at $\omega_i^* \propto N_k^i N_{k'}^i / (\alpha N_k^i + \beta N_{k'}^i)$. Harmonic is the $\alpha = \beta$ special case.

Theorem 3 is unconditional in the counts – a property that distinguishes harmonic from the min-count heuristic of [1], which has no such guarantee. Theorem 4 provides the oracle benchmark; in synthetic experiments substituting the true α, β into ω^* yields variance within $\pm 2\%$ of harmonic. Complete proofs and additional sensitivity analyses are provided in the supplementary material.

Limits of the symmetric guarantee. The unconditional bound in Theorem 3 concerns $V_k + V_{k'}$. The asymmetric analogue $\alpha V_k(\omega^h) + \beta V_{k'}(\omega^h) \leq \alpha V_k(1) + \beta V_{k'}(1)$ does *not* hold unconditionally: a closed-form $n = 2$ counterexample with $(N_k^1, N_{k'}^1) = (1, 100)$, $(N_k^2, N_{k'}^2) = (100, 1)$ and $(\alpha, \beta) = (0.01, 1)$ has harmonic strictly worse than uniform on the asymmetric surrogate. The configurations under which this fails are degenerate (each cell informative at exactly one position); we have not encountered them in any realistic count distribution we tested. The ω^* in Theorem 4 remains the exact oracle for all (α, β) .

Table 1: Plug-in SE vs. click-resample SE across KDD subsamples, and ARE(harmonic, uniform) at each scale. Click resample only computed at small f for runtime; agreement is within 1–2% where measured. ARE grows with data volume because heavy-tail mass becomes more pronounced at larger scales.

f	pairs	$\widehat{\text{SE}}_U^{\text{plug}}$	$\widehat{\text{SE}}_U^{\text{boot}}$	$\widehat{\text{SE}}_H^{\text{plug}}$	ARE
10^{-3}	4.3K	0.131	0.132	0.111	1.66
10^{-2}	54K	0.039	0.038	0.028	2.21
10^{-1}	542K	0.012	–	0.007	3.07
1.0	4.61M	0.0042	–	0.0020	4.53

3 Plug-in Asymptotic Variance: the Right Instrument

$\tilde{V}$ in closed form. With $A = \sum_i \omega_i \theta_k^i$, $B = \sum_i \omega_i \theta_{k'}^i$, and clicks at k and k' conditionally independent given counts, the delta method gives

$$\tilde{V}(\omega) = \frac{\sum_i \omega_i^2 \theta_k^i (1 - \theta_k^i) / N_k^i}{A^2} + \frac{\sum_i \omega_i^2 \theta_{k'}^i (1 - \theta_{k'}^i) / N_{k'}^i}{B^2}, \quad (2)$$

$\text{SE}(\hat{R}) = |\hat{R}| \sqrt{\tilde{V}(\omega)}$. Plugging in $\hat{\theta}_i$ (with the shrinkage estimator $\hat{\theta}_i = (C_i + \bar{\theta}_k) / (N_i + 1)$, $\bar{\theta}_k$ the marginal position- k CTR) yields the textbook self-normalised IS variance estimator [11–13].

ARE. For schemes a, b , $\text{ARE}(a, b) = \tilde{V}(\omega_b) / \tilde{V}(\omega_a)$: scheme b needs $\text{ARE}(a, b)$ times the data of a for equal precision. $\tilde{V}$ scales as $1/\text{ESS}(\omega)$, with Kong’s effective sample size [11] $\text{ESS}(\omega) = (\sum_i \omega_i)^2 / \sum_i \omega_i^2$.

Why pair-bootstrap is misaligned here. A naive bootstrap that re-samples (q, d) pairs estimates a different quantity than the conditional-on-counts efficiency we care about, for three reasons. (i) Averaging over $\sim 60\%$ near-singleton pairs under uniform produces a smooth resampling distribution whose tightness reflects averaging-of-noise rather than estimator precision. (ii) Pair resampling perturbs the high-information cells that drive any inverse-variance estimator, so concentrated weight schemes are systematically penalised by the resampling design, not by their statistical behaviour. (iii) Pair-bootstrap entangles estimator efficiency with model fit; $\tilde{V}$ separates them (Section 4). The appropriate empirical baseline against which to validate the plug-in is the click-resample (parametric Bernoulli bootstrap conditional on counts), which we adopt for calibration.

Calibration and stability at scale. Table 1 reports plug-in SE versus empirical click-resample SE across KDD sample fractions, together with ARE(harmonic, uniform) at each scale. Because both $\tilde{V}$ and click-resample SE share the conditional Bernoulli + cross-cell-independence assumption, their agreement is primarily an implementation-correctness check; a stronger probe of the independence assumption would be a query-cluster bootstrap, which is sensitive to Cochran- Q -style heterogeneity. We report pair-bootstrap numbers in supplementary material. The ARE grows with sample size ($1.66 \rightarrow 4.53$) because the heavy-tailed structure of the count distribution becomes more pronounced as more data is included rather than averaged out.

Table 2: Plug-in delta-method asymptotic variance $\tilde{V}$, SE of $\hat{R}$, ARE vs. uniform, and ESS on full KDD Cup 2012 Track 2 ($n_{\text{pairs}} = 4.61\text{M}$). Harmonic, min, and the count-only oracle all reach ARE ≈ 4.5 . The full- θ oracle is pathological because per-cell $\hat{\theta}_i$ is too noisy at this scale.

Scheme	$\hat{R}$	SE	ARE	ESS
Uniform	1.683	4.24e-3	1.00	4.61M
Min [1]	1.712	2.05e-3	4.41	2,526
Harmonic	1.710	2.02e-3	4.53	1,849
Count-oracle (0.5,1)	1.715	2.02e-3	4.57	1,760
θ -aware oracle	2.343	0.148	0.002	1.1

Table 3: Stratified ARE(harmonic, uniform) by $N_1 + N_2$ tertile on full KDD (the bottom quartile boundary collapses into the lowest tertile because singletons dominate). Schemes are tied on the noise-dominated strata; harmonic dominates $\sim 5\times$ on the high-information stratum where the actual estimation signal lives.

$N_1 + N_2$ tertile	n pairs	$\tilde{V}_U$	ARE(h, u)
Lowest	1.99M	2.3e-5	1.02
Middle-low	1.38M	1.9e-5	1.09
Highest	1.25M	7.6e-6	4.98

4 Real-Log Results: KDD Cup 2012 Track 2

Data. Full release: 149.6M raw rows expanding to 235.6M impressions, 63.8M unique (q, ad, pos) cells, 8.2M clicks (CTR 3.49%). Pair (1, 2) interventional set: 4.61M (q, ad) pairs, 132.3M impressions, median pair size 4, max 2.47M; 60.7% of pairs have $N_1 + N_2 \leq 5$.

Headline (Table 2). ARE(harmonic, uniform) = 4.53: **uniform requires $\sim 4.5\times$ the impressions of harmonic for matching precision** on the relevance-agnostic surrogate. The min-count heuristic [1] achieves 4.41 on the same surrogate (-2.7% relative); harmonic and min are essentially tied empirically on KDD; harmonic’s contribution is the unconditional guarantee (Theorem 3) and the parameter-free derivation, not a large empirical margin. The count-only oracle (4.57) is the ceiling within the surrogate. The θ -aware oracle, which would be optimal under Aitken’s theorem with known r_i , plugs in $\hat{r}_i$ that is too noisy on the singleton-dominated tail and collapses to ARE = 0.002: a substantive finding rather than a side note—it shows that count-only weighting is robust precisely where plug-in IVW fails.

Asymmetric surrogate (empirical). With $\hat{p}_1 = 0.0501$, $\hat{p}_2 = 0.0256$ estimated from the marginal CTRs, $\alpha = 18.97$, $\beta = 38.08$, $\alpha/\beta \approx 0.50$. Harmonic reduces $\alpha V_k + \beta V_{k'}$ by a factor of 4.75 versus uniform—essentially identical to the symmetric reduction of 4.84. The asymmetric counterexample of Section 2 thus does not materialise on KDD’s count distribution.

Stratified ARE (Table 3). Schemes tie on noise singletons (where there is nothing to estimate) and harmonic dominates uniform by $5\times$ on the high-information stratum. The gain comes from *not* treating singletons as equal to million-impression cells.

Table 4: Count-only weighting schemes compared on full KDD via plug-in $\tilde{V}$. Schemes that concentrate more aggressively give lower $\tilde{V}$ but smaller ESS. Harmonic is the inverse-variance optimum among count-only schemes (Theorem 2).

Scheme	$\hat{R}$	SE	ARE	ESS
Uniform	1.683	4.24e-3	1.00	4.61M
$\log(1 + \min(N_k, N_{k'}))$	1.651	3.18e-3	1.71	3.17M
$\sqrt{N_k N_{k'}}$	1.706	2.28e-3	3.55	1,246
$\min(N_k, N_{k'})$	1.712	2.05e-3	4.41	2,526
Harmonic	1.710	2.02e-3	4.53	1,849
Harmonic, cap@99th pct	1.606	2.40e-3	2.83	765K

Weight-cap robustness. The harmonic ESS of 1,849 out of 4.61M pairs invites a concern that the efficiency gain rests on a few extreme cells, but the small ESS reflects deliberate concentration on the pairs containing nearly all statistical information in this heavy-tailed setting rather than instability of the estimator. Capping ω_i^h at the 99.99th percentile (threshold 2,079; 462 pairs capped) lifts ESS to 5.2×10^4 and still gives ARE = 4.12; at the 99th percentile, ARE = 2.83 (ESS 7.6×10^5); even at the 90th, ARE = 1.86. Harmonic beats uniform robustly across capping levels.

Comparison of count-only weighting schemes. Table 4 compares harmonic to the natural alternatives on full KDD. Among count-only weights, harmonic is best on $\tilde{V}$. Log-counting is too flat (treats singletons nearly as well as million-impression cells); the geometric mean and min-count concentrate more aggressively but neither matches the inverse-variance optimum. Capping at the 99th percentile preserves a $2.83\times$ advantage but shifts the estimand ($\hat{R} : 1.71 \rightarrow 1.61$), consistent with the estimand discussion below.

Estimand under PBM mis-specification. PBM is a known approximation; on real logs different weight schemes estimate different functionals of cell-specific ratios. Specifically, the population limit of $\hat{R}$ under any weight ω is

$$R_\omega = \frac{\sum_i \omega_i \theta_k^i}{\sum_i \omega_i \theta_{k'}^i},$$

which equals $p_k/p_{k'}$ only when $\theta_k^i/\theta_{k'}^i$ is the same constant across cells—i.e. when PBM holds exactly. Under mis-specification each scheme thus targets a different weighted average of the cell-specific ratios $\rho_i = \theta_k^i/\theta_{k'}^i$ (uniform $\rightarrow 1.68$, harmonic $\rightarrow 1.71$, θ -aware oracle $\rightarrow 2.34$ on KDD). The plug-in $\tilde{V}$ comparison is a precision statement at fixed ω —a model-based efficiency comparison—and does *not* claim that one R_ω is the “correct” position-bias ratio. Choosing ω thus involves a precision/estimand trade-off; harmonic’s attractive property is that, among count-only schemes, it concentrates on high-information cells (where ρ_i is estimated most precisely) without depending on noisy plug-in $\hat{\theta}_i$.

Quantifying PBM mis-specification (Cochran’s Q). PBM is a known approximation, not a literal model – prior work [7, 15, 16] has long established that real logs violate it. The relevant question is how *large* the deviation is. Cochran’s Q on the 16,132 high-information pairs (≥ 5 clicks at each of positions 1 and 2) gives $Q = 6.17\times 10^4$, $\text{df} = 16,131$, $Q/\text{df} = 3.83$: per-cell odds-ratio variability is $\sim 3.8\times$

Table 5: MSE decomposition under three explicit PBM violations (anchor imbalance 50:1, split 80/20, 5,000 trials). Count-only schemes share comparable bias² within each model; harmonic achieves the lowest variance, hence the lowest MSE.

Model	Scheme	MSE	Variance	Bias ²
Trust	Uniform	0.00487	0.00486	1.0e-5
	Min	0.00298	0.00297	2.0e-5
	Harmonic	0.00272	0.00272	0.0
Cascade	Uniform	0.01376	0.00276	0.01100
	Min	0.01242	0.00171	0.01071
	Harmonic	0.01252	0.00147	0.01105
Pos.-dep.	Uniform	0.04086	0.03973	0.00113
	Min	0.03896	0.03770	0.00127
	Harmonic	0.03724	0.03613	0.00111

what binomial sampling alone would produce, well outside Poisson dispersion. This magnitude is the quantitative driver of the estimand drift discussed in Paragraph 4; plug-in $\tilde{V}$ keeps the efficiency comparison well-posed by working at fixed ω .

5 Synthetic and Yahoo-LTR Validation

To validate the ARE prediction in settings with ground truth, we run two controlled experiments.

Anchor-item imbalance sweep (synthetic). Following [1]: $M=10$ positions, $p_k = k^{-\eta}$ with $\eta = 1$, AdjacentChain estimator. A fraction s of documents are “anchors” with heavy traffic at one position ($N \sim \text{Unif}(50r, 200r)$) and light traffic at the other ($N \sim \text{Unif}(1, 51)$); the remaining $1 - s$ are balanced ($N \sim \text{Unif}(25, 75)$ at both positions); $s \in \{0.5, \dots, 0.95\}$, $r \in \{1, \dots, 100\}$, 500 trials per cell. At $s = 0.80$, $r = 100$ harmonic reduces variance by 43.6% vs. uniform and 5% vs. min; reduction grows to $\sim 54\%$ at $s = 0.90$, $r = 50$. Across the full (s, r) grid the data-equivalence factor $\text{DEF}(s, r) = b_{\text{uni}}(s, r)/b_{\text{harm}}(s, r)$ – the ratio of base impressions per document at which each scheme first hits a fixed MSE threshold – has median 1.31, mean 1.30, max 1.94. Highly imbalanced cells achieve up to $2\times$ data savings; near-balanced cells ($r = 1$) saturate at $\text{DEF} \approx 1$.

Bias comparability under PBM violations. We additionally simulate three explicit PBM violation models – trust bias, cascade, and position-dependent relevance [1, 6] – at imbalance ratio 50:1, split 80/20, 5,000 trials. Across all three models the count-only schemes (uniform, min, harmonic) exhibit *comparable* bias², so harmonic’s lower variance translates directly into lower MSE (Table 5). Under cascade, where history-dependent examination introduces a substantial structural mismatch (bias² ≈ 0.011 for every ratio-unbiased scheme), the variance edge is preserved but the practical MSE gain shrinks because bias dominates.

Yahoo-LTR semi-synthetic. Same simulator with graded relevance from the Yahoo Learning to Rank Challenge Set 1 test split (6,936 queries; relevance grades $g \in \{0, \dots, 4\}$ mapped to $r(q, d) = (2^g - 1)/15$ following NDCG; click model $P(C = 1 \mid q, d, k) = p_k r(q, d)$; 200 trials). Harmonic reduces variance by $\sim 40\%$ vs. uniform at imbalance $r = 50$ ($\text{Var } 1.5 \times 10^{-4} \rightarrow 8.7 \times 10^{-5}$). Aggressive clipping

trades variance for bias and yields $\sim 458\times$ higher MSE, confirming that variance-reduction without bias control is not viable.

Production-style traffic simulation. $K=5$ concurrent rankers; a default ranker shows each document at its natural position, treatment rankers swap adjacent pairs with probability 0.5. Under uniform traffic ($5 \times 20\%$) only 0.4% of adjacent pairs exceed impression ratio 10 and harmonic’s MSE reduction is 6.3%; under production traffic ($80\% + 4 \times 5\%$), 16.7% of pairs exceed ratio 10 and harmonic’s MSE reduction grows to **17.2%**, confirming the gain scales directly with traffic-induced imbalance.

6 Related Work

Intervention harvesting was introduced by [1] with the uniform and min-count variants we compare against; [15] jointly estimates relevance and propensities via EM. Self-normalised weighted ratio estimators have a long history in importance sampling and survey statistics: Owen [12] §9.2 derives the asymptotic variance we use; Kong [11] introduces ESS; Hesterberg [4] analyses heterogeneous-weight variance. Inverse-variance weighting traces to Aitken [2] and is canonical in meta-analysis [5, 8]. The closest analogue in counterfactual learning is the self-normalised IPS estimator SNIPS [13], which applies the same $\tilde{V}$ framework to off-policy evaluation in bandits; [3] analyses why weight heterogeneity, not sample count, governs effective precision on logged feedback. PBM violations in real logs are documented in [15, 16] and reviewed in [7].

Why count-only PBM weighting is still worth studying. Given that PBM is approximate (Section 4), one might prefer EM-style joint estimation of relevance and propensities as in [15], which has bias-variance trade-offs of its own (EM is sensitive to initialisation and model choice; intervention harvesting is one-shot and model-internal). The two approaches are complementary: intervention harvesting is a clean building block whose variance properties we characterise here, and a relevance-agnostic weight scheme like harmonic is precisely the right input to an EM M-step that re-uses count statistics. A direct empirical comparison against an EM estimator is left to the regular-paper version.

7 Conclusion

We frame harmonic weighting as the inverse-variance optimum of an explicit *relevance-agnostic count-only surrogate* of the delta-method variance under PBM (Theorems 2–4). This is honest about the cost of dropping cell relevance from the actual Bernoulli variance, which the θ -aware oracle exposes as plug-in infeasibility on singleton-dominated data ($\text{ARE} = 0.002$). The plug-in delta-method variance $\tilde{V}$ (Eq. 2) is the natural diagnostic at fixed ω ; on full KDD Cup 2012 Track 2 it yields $\text{ARE}(\text{harmonic}, \text{uniform}) = 4.53$ on the surrogate, rising to $\sim 5\times$ on information-bearing cells, with harmonic and the min heuristic essentially tied empirically (4.53 vs. 4.41). Under known PBM mis-specification (Cochran- $Q/\text{df} = 3.83$), different schemes target different weighted averages of cell-specific ratios; choosing ω is a precision/estimand trade-off, not a search for “the” position-bias ratio. Future work: shrinkage variants of the θ -aware oracle (Theorem 4 with θ_i regularised); empirical comparison against EM-style joint estimation [15]; unconditional out-of-policy variance bounds; click models beyond PBM [7, 16].

GenAI Usage Disclosure

The authors used a large-language-model assistant for code generation (KDD aggregation, plug-in variance script, LaTeX table formatting), and for copy-editing. All experimental design, mathematical derivations (Theorems 2–4 and Eq. (2)), interpretation, citations, and final wording were authored and verified by the authors. No LLM-generated content was used without human review. Full code, scripts, and data-processing pipelines are provided as supplementary material.

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